

PROJECT REPORT

Advanced Packet Sniffer and ARP Spoofing Detector (Using Python and Scapy)

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1. Introduction:

This project is a network security tool developed using Python and the Scapy library. It captures and analyzes network packets in real time and detects ARP spoofing attacks.

2. Objectives:

- Capture ARP packets from network traffic
- Monitor IP-MAC address mapping
- Detect ARP spoofing attacks
- Generate real-time alerts
- Log detected events for analysis

3. Technologies Used:

- Python
- Scapy Library
- Kali Linux

4. Working Principle:

The tool captures ARP packets, extracts IP and MAC addresses, and maintains a mapping table. If the same IP is associated with a different MAC address, it detects a spoofing attempt and alerts the user. All events are logged in a CSV file.

5. Features:

- Real-time packet sniffing
- ARP packet filtering
- IP-MAC mapping system
- ARP spoofing detection
- Alert generation system
- CSV logging system
- Timestamp-based monitoring

6. Upgrades from Basic Version:

Feature	Basic Version	Upgraded Version
Packet Sniffing	✓	✓
ARP Monitoring	✓	✓
Spoof Detection	✓	✓
Alert System	✗	✓
Logging System	✗	✓
Timestamp	✗	✓
Duplicate Alert Control	✗	✓

7. Output:

The tool successfully detects new devices and identifies ARP spoofing attempts.

```
(kali@kali)-[~/packet_sniffer_project]
$ sudo python3 arp_detector.py
[sudo] password for kali:
Advanced Packet Sniffer Started...
Monitoring ARP traffic for spoofing detection...
Press CTRL + C to stop.

[2026-04-08 11:17:20] [NEW DEVICE] 10.131.238.251 → ca:53:b9:62:77:4e
[2026-04-08 11:18:02] [NEW DEVICE] 10.131.238.252 → 60:ff:9e:50:71:3e
[2026-04-08 11:18:35] [NEW DEVICE] 10.131.238.1 → 00:0c:29:b5:5c:53

[2026-04-08 11:18:41] ▲ ARP SPOOFING DETECTED!
IP Address : 10.131.238.252
Old MAC    : 60:ff:9e:50:71:3e
New MAC    : 00:0c:29:b5:5c:53
```

Figure 1: ARP Spoofing Detection in Terminal Output

```
(kali@kali)-[~/packet_sniffer_project]
$ ls
arp_detector.py  arp_log.csv

(kali@kali)-[~/packet_sniffer_project]
$ column -s, -t arp_log.csv
Time           Event           IP Address      Old MAC         New MAC
2026-04-08 11:17:20 New Device      10.131.238.251 -              ca:53:b9:62:77:4e
2026-04-08 11:18:02 New Device      10.131.238.252 -              60:ff:9e:50:71:3e
2026-04-08 11:18:35 New Device      10.131.238.1   -              00:0c:29:b5:5c:53
2026-04-08 11:18:41 ARP Spoofing Detected 10.131.238.252 60:ff:9e:50:71:3e 00:0c:29:b5:5c:53
```

Figure 2: Logged ARP Events in CSV File

8. Ethical Considerations:

This project was developed and tested in a controlled environment using a personal network. Unauthorized packet sniffing or monitoring on public or private networks without proper permission may be illegal and unethical.

The tool is intended only for educational and security research purposes. It should be used responsibly to understand network vulnerabilities and improve security practices.

9. Conclusion:

The project successfully demonstrates how ARP spoofing attacks can be detected using packet sniffing techniques. The upgraded version improves detection accuracy by adding real-time alerts, logging, and structured monitoring.

The system effectively identifies changes in IP-MAC mappings and provides immediate alerts, making it useful for basic network security monitoring. The logging feature also helps in maintaining records for further analysis.

This project provides practical knowledge of network security concepts and demonstrates how ARP-based attacks can be identified in a controlled environment. The system can be further enhanced with advanced analysis features in future work.